Practical 3: Iris Dataset using Decision Tree.

In [11]:
```
import numpy as np
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
import matplotlib.pyplot as plt
from sklearn.tree import plot_tree
```

In [3]:
```
iris = load_iris()
```

In [ ]:
```
X_train, X_test, y_train, y_test = train_test_split(
    iris.data,iris.target, test_size=0.3, random_state=0
    )
```

In [6]:
```
dtc=DecisionTreeClassifier(random_state=32)

dtc.fit(X_train,y_train)
```

Out[6]:

▾ DecisionTreeClassifier

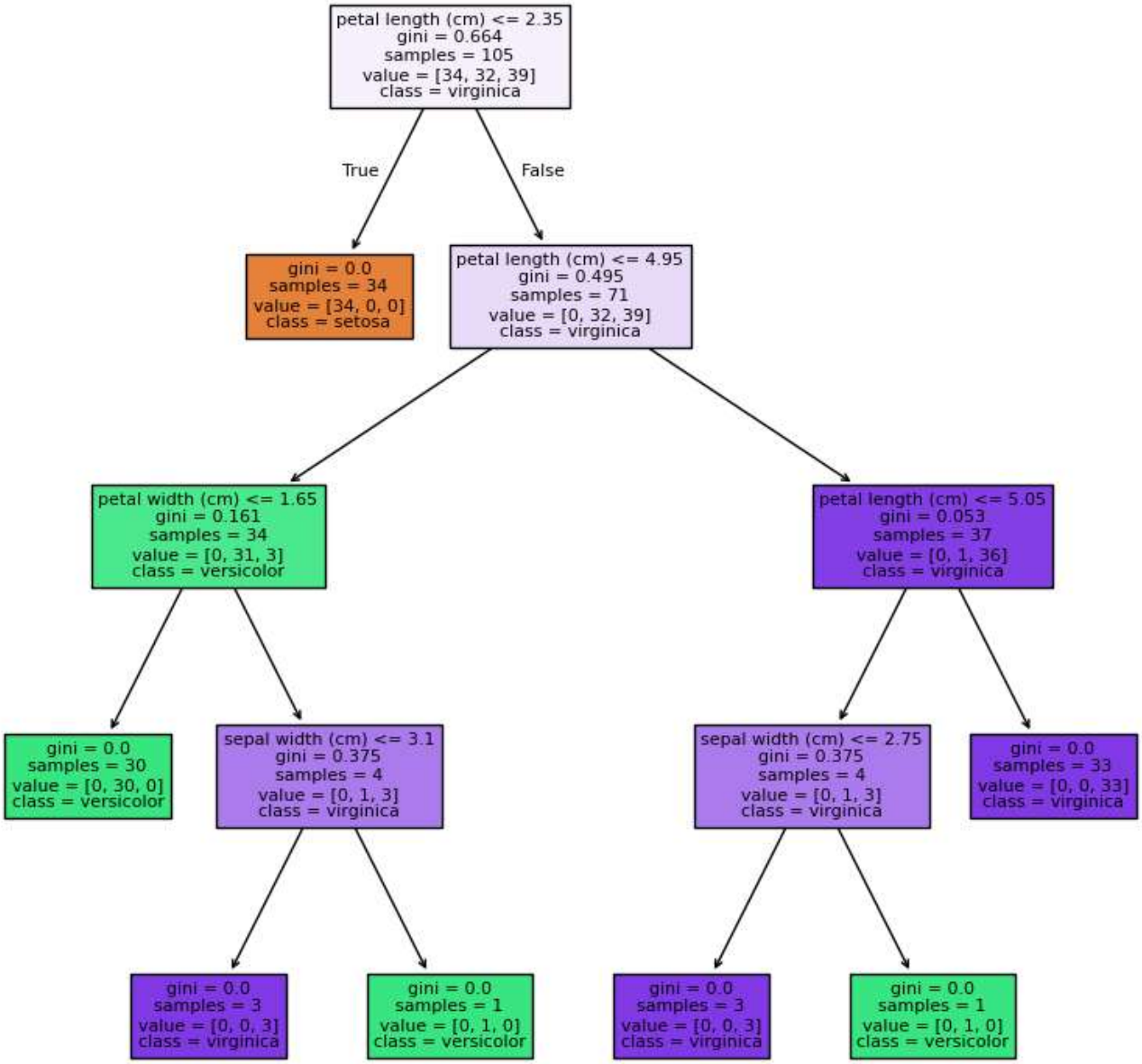
▶ Parameters

In [7]:
```
y_pred=dtc.predict(X_test)
```

In [10]:
```
accuracy=(y_test==y_pred).mean()
print("Accuracy:", round(accuracy*100,2), "%")
```

```
Accuracy: 97.78 %
```

In [16]:
```
plt.figure(figsize=(10,10))
plot_tree(dtc, filled=True, feature_names=iris.feature_names,
class_names=iris.target_names)
plt.show()
```


petal length (cm) <= 2.35
gini = 0.664
samples = 105
value = [34, 32, 39]
class = virginica
True
False
gini = 0.0
samples = 34
value = [34, 0, 0]
class = setosa
petal length (cm) <= 4.95
gini = 0.495
samples = 71
value = [0, 32, 39]
class = virginica
petal width (cm) <= 1.65
gini = 0.161
samples = 34
value = [0, 31, 3]
class = versicolor
petal length (cm) <= 5.05
gini = 0.053
samples = 37
value = [0, 1, 36]
class = virginica
gini = 0.0
samples = 30
value = [0, 30, 0]
class = versicolor
sepal width (cm) <= 3.1
gini = 0.375
samples = 4
value = [0, 1, 3]
class = virginica
sepal width (cm) <= 2.75
gini = 0.375
samples = 4
value = [0, 1, 3]
class = virginica
gini = 0.0
samples = 33
value = [0, 0, 33]
class = virginica
gini = 0.0
samples = 3
value = [0, 0, 3]
class = virginica
gini = 0.0
samples = 1
value = [0, 1, 0]
class = versicolor
gini = 0.0
samples = 3
value = [0, 0, 3]
class = virginica
gini = 0.0
samples = 1
value = [0, 1, 0]
class = versicolor